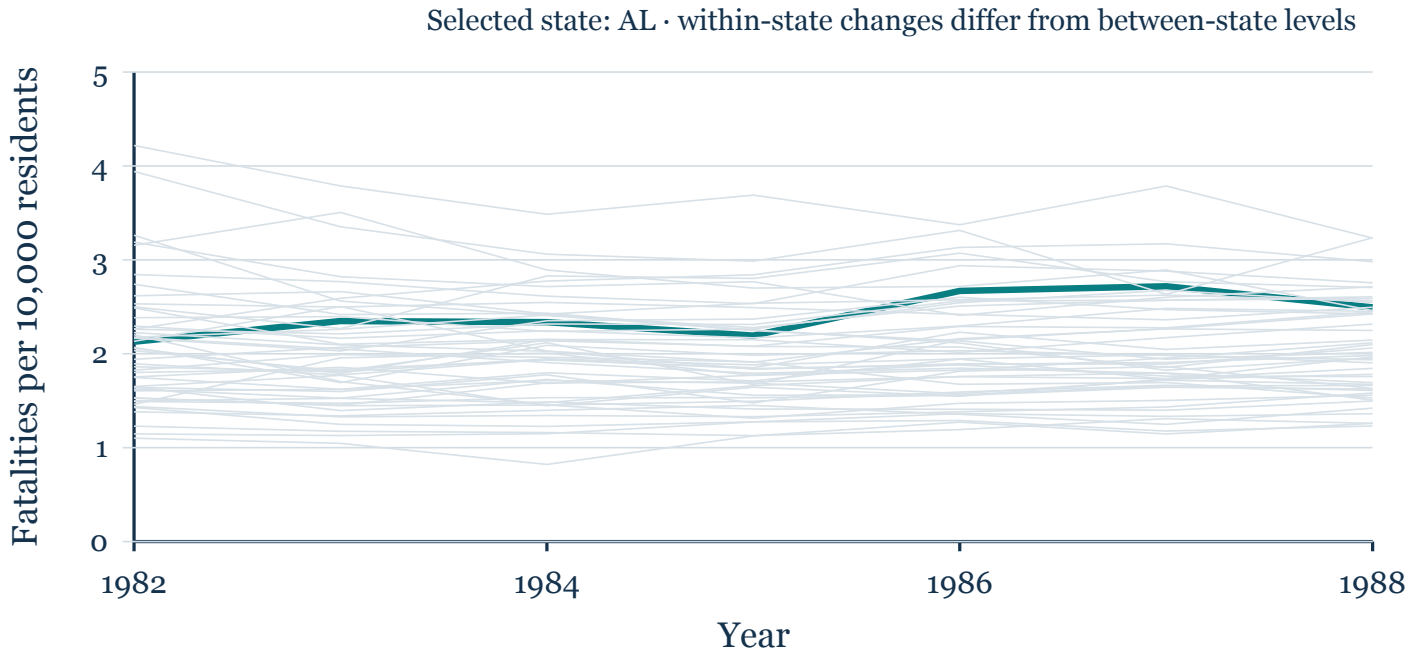
Can repeated observations improve a policy comparison?

Use changes within states to study alcohol taxes and traffic fatalities.



Observed state fatality rates, 1982–1988. The control selects a state; pale paths show the remaining states.

A panel has identities as well as dates

Repeated cross sections do not give the same within-unit comparison.

> One row is a state in a year.

> Before estimating, verify that the state-year pair uniquely identifies rows and inspect the coverage.

Write a stable omitted effect explicitly

An unobserved unit component may be related to policy.

$$Y_{it} = \beta X_{it} + \alpha_i + \lambda_t + u_{it}$$

α_i: stable state component; λ_t: common year shift.

u_{it}: remaining state-year error.

Follow one unit across two dates

Subtracting the same unit's outcomes removes its stable intercept.

> $(X_{i1}, X_{i2}) = (2, 4), \quad (Y_{i1}, Y_{i2}) = (10, 14)$

> $\Delta X_i = 2, \quad \Delta Y_i = 4, \quad \alpha_i - \alpha_i = 0$

First differencing removes a stable unit component

Subtract the earlier equation from the later equation.

> $$\Delta Y_i = \beta \Delta X_i + \Delta \lambda + \Delta u_i$$

A two-period athlete example uses the same algebra

State what disappears and what the comparison still needs.

> $GPA_{it} = \beta\, season_{it} + \alpha_i + \lambda_t + u_{it}$

> $\Delta GPA_i = \beta \Delta season_i + \Delta\lambda + \Delta u_i$

> GPA: grade point average; season: a zero-or-one indicator.

First differences

Try the comparison before opening the explanation.

Practice

In $GPA_{it} = \beta\, season_{it} + \alpha_i + \lambda_t + u_{it}$ for two semesters, write the differenced equation and say what disappears.

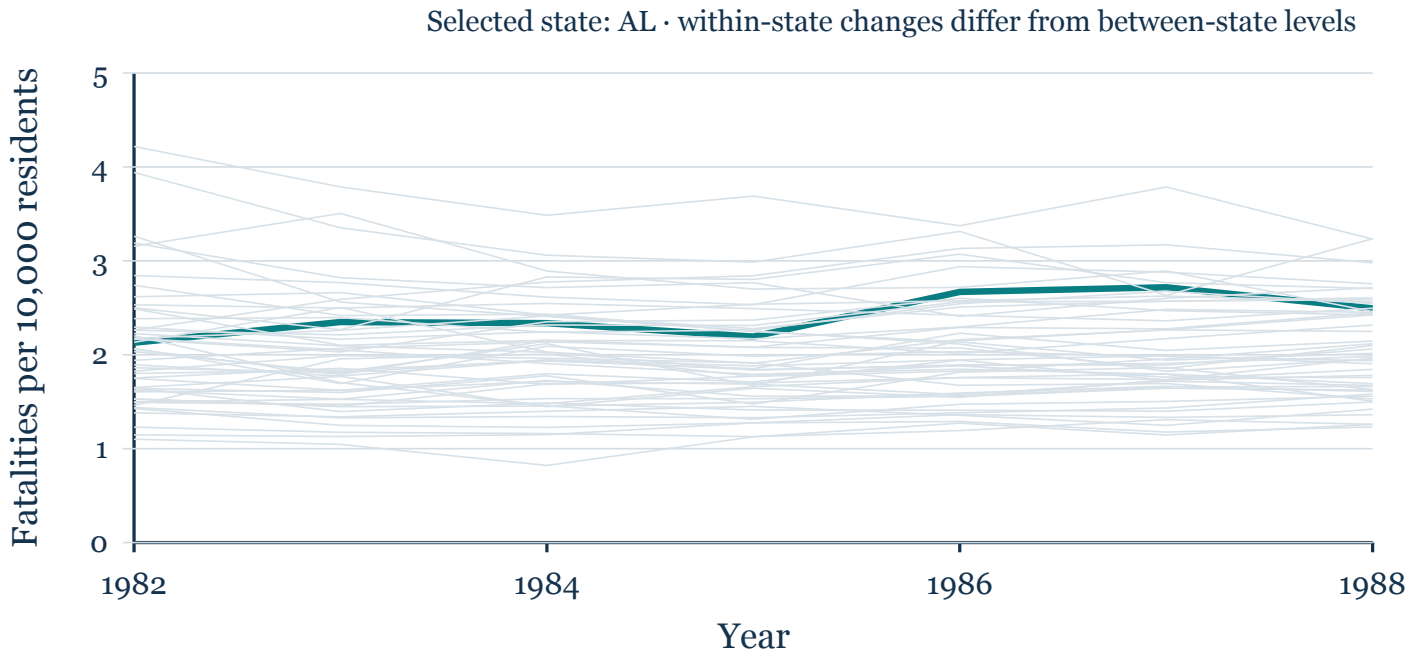
First differences: explanation

Follow the calculation and interpret its units.

> $\Delta GPA_i = \beta \Delta season_i + \Delta\lambda + \Delta u_i$. The time-invariant α_i disappears, as do coefficients on time-invariant regressors. Semester effects do not disappear; with two periods they become a common intercept. Time-varying confounding can remain.

Demeaning uses each unit's own average

Fixed effects remove the same stable component over several periods.



Observed state fatality rates, 1982–1988. The control selects a state; pale paths show the remaining states.

$$X_i = (2, 4) \Rightarrow X_i - \overline{X}_i = (-1, 1)$$

$$Y_i = (10, 14) \Rightarrow Y_i - \overline{Y}_i = (-2, 2)$$

$$Y_{it} - \overline{Y}_i = \beta(X_{it} - \overline{X}_i) + (\lambda_t - \overline{\lambda}) + (u_{it} - \overline{u}_i)$$

Two-period fixed effects and first differences share a slope identity

The identity requires the same two-period sample and specification.

> With two observations per unit, demeaning produces one half of the change and its negative.

> Our application compares a two-period change from 1982 to 1988 with a fixed-effects fit using all seven years.

Read pooled and within-state associations

Removing stable differences can change the sign.

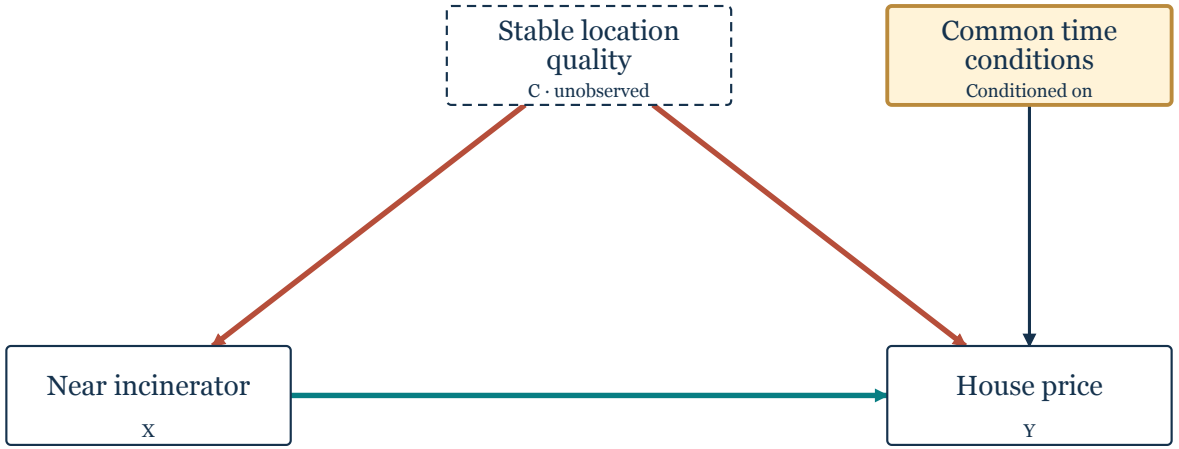
```
. xtreg fatality beertax i.year, fe vce(cluster state)

------------------------------------------------------------------------------
             |               Robust
    fatality | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     beertax |  -.6399799   .3570783    -1.79   0.080    -1.358329     .078369
             |
       _cons |    2.42847   .2016884    12.04   0.000     2.022725    2.834215
-------------+----------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

Use year effects for common shocks

They do not remove state-specific changes.



1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

> Year effects: common date shifts. State effects: stable additive state differences.

> A location common cause is not removed by a common calendar control.

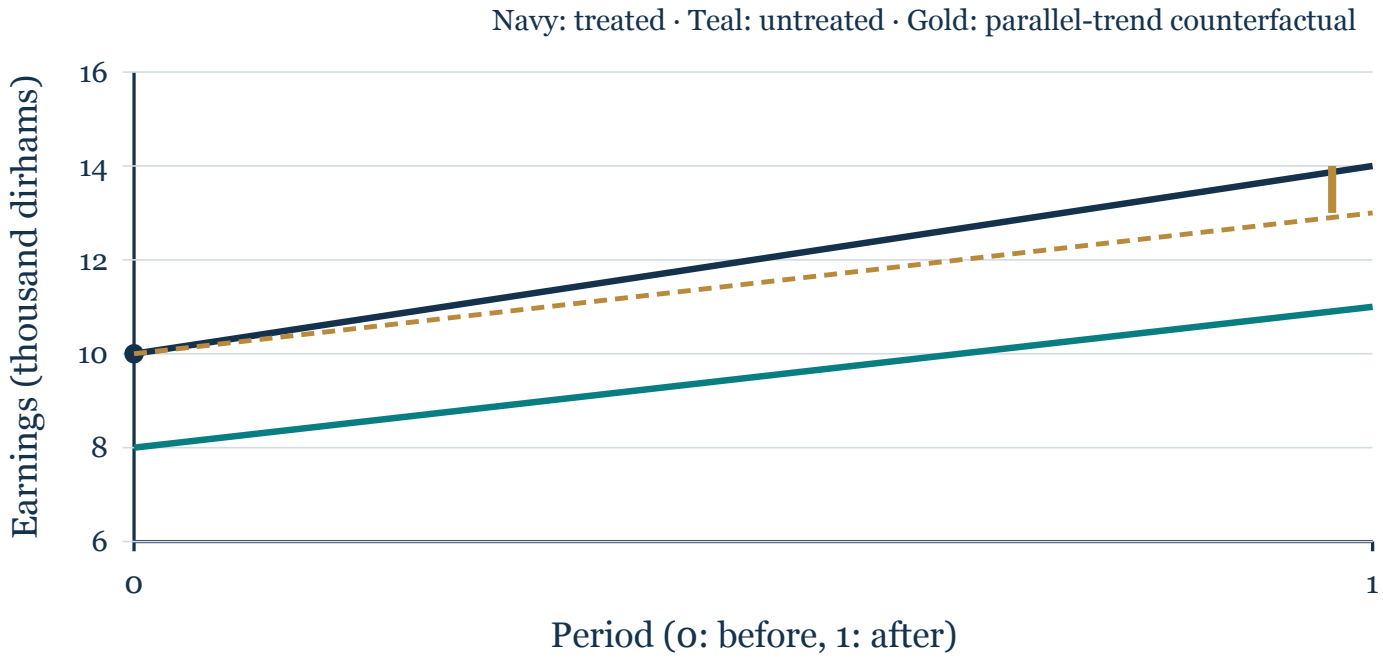
State the fixed-effects exogeneity condition

Transforming variables does not make their remaining changes random.

> $E(u_{it} \mid X_{i1}, \ldots, X_{iT}, \alpha_i) = 0$ as a sufficient panel condition

Build a difference-in-differences comparison

Calculate the observed contrast before assigning it a causal meaning.



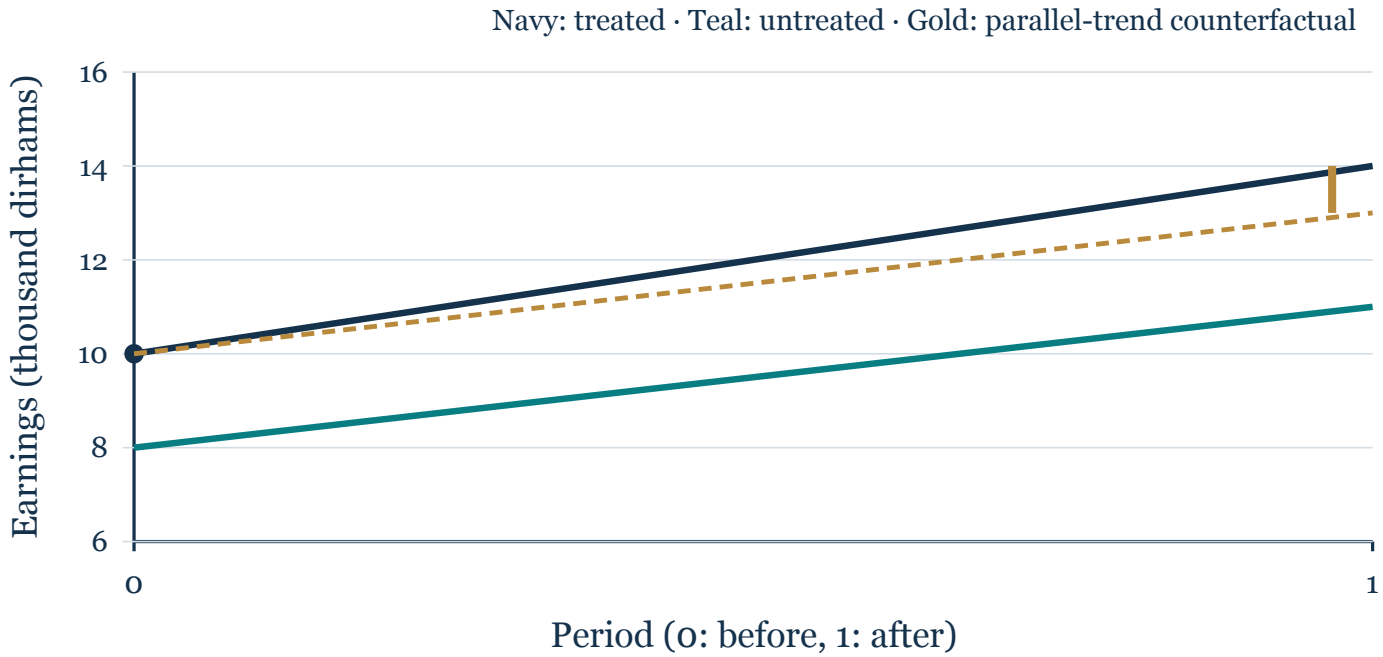
Constructed earnings example, in thousand dirhams per month. The dashed counterfactual assumes the untreated group's three-unit change.

> Treated: $10 \to 14$; comparison: $8 \to 11$

> $\widehat{\tau} = (14 - 10) - (11 - 8) = 1$ thousand dirhams

Parallel trends is about unobserved untreated changes

It is not a requirement that the groups begin at the same level.



Constructed earnings example, in thousand dirhams per month. The dashed counterfactual assumes the untreated group's three-unit change.

- Counterfactual treated post mean under parallel untreated trends: $10 + (11 - 8) = 13$
- Effect under the maintained assumptions: $14 - 13 = 1$
- The unobserved untreated change is the identifying claim.

No anticipation and stable comparison units also matter

Define when treatment can begin to affect behaviour.

> No anticipation means that future treatment does not already alter the pre-treatment outcomes used in the comparison.

> Also examine whether group composition changes and whether treatment affects the comparison group's outcomes through spillovers.

DiD arithmetic

Try the comparison before opening the explanation.

Practice

Treated mean outcomes rise from 10 to 14; comparison means rise from 8 to 11. Compute DiD and state its identifying assumption.

DiD arithmetic: explanation

Follow the calculation and interpret its units.

> DiD $= (14 - 10) - (11 - 8) = 1$. A causal interpretation needs parallel untreated trends, no anticipation and suitable composition/spillover assumptions. Similar baseline levels are not required; the key assumption concerns counterfactual changes.

The interaction in a two-by-two DiD equation has a specific meaning

Its interpretation follows the four group-time means.

> $Y = a + bG + cP + \tau(GP) + u$

> G: treated-group membership; P: post period; GP: treated-post exposure.

Cluster inference for repeated units

Errors within a state can remain related after fixed effects.

> State fixed effects remove a stable mean component; they do not force the remaining errors in different years to be independent.

> The policy assignment and shared shocks help determine the appropriate clustering unit.

Cluster choice

Try the comparison before opening the explanation.

Practice

A policy is assigned to states and repeated outcomes are observed for many districts over years. Why is clustering by district alone potentially insufficient?

Cluster choice: explanation

Follow the calculation and interpret its units.

> Errors and policy exposure may be shared across districts within a state. Inference should reflect the assignment and dependence structure; state clustering is a starting point when states are independent sampling/assignment clusters. Few states require special care, and a universal cluster-count threshold is not a theorem.

Interpret a feasible tax difference

Use the fitted contrast and preserve the outcome units.

```
. lincom .10*beertax

------------------------------------------------------------------------------
    fatality | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   -.063998   .0357078    -1.79   0.080    -.1358329    .0078369
------------------------------------------------------------------------------
```

Inspect the 1982-to-1988 changes separately

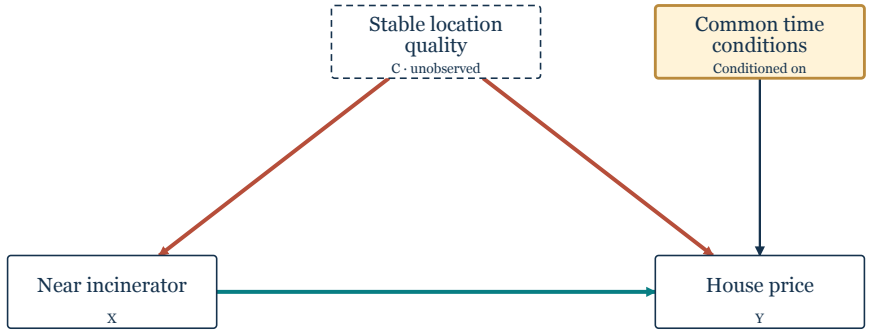
A longer change can use different variation from annual fixed effects.

```
. regress dy dx, vce(robust)

------------------------------------------------------------------------------
             |               Robust
          dy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
          dx |  -1.040973   .3550061    -2.93   0.005    -1.755563   -.3263823
       _cons |  -.0720371   .0653552    -1.10   0.276    -.2035903    .0595161
------------------------------------------------------------------------------
```

DAG practice: pooling is not within-location identification

Trace the routes before choosing a control.



1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

In the incinerator example, common time conditions affect prices while stable location quality affects proximity and prices. Do year indicators close the location route? If proximity never changes, what happens to its level coefficient under unit fixed effects, and what extra comparison could support a DiD question?

DAG practice: pooling is not within-location identification: explanation

Explain what conditioning changes in this stated graph.

> Year indicators do not block $X \leftarrow C \rightarrow Y$ through stable location quality. Unit fixed effects can cancel an additive stable location component, but they also absorb a proximity indicator that never changes within a unit. Its separate level coefficient is then unavailable. A proximity-by-post-period interaction can vary across groups and dates. Interpreting that interaction as a DiD effect requires a well-defined intervention and parallel untreated trends, along with the other stated comparison assumptions. Neither pooling nor the causal diagram alone establishes those assumptions.

Distinguish within-state and endpoint comparisons

Reproduce the evidence, then explain its economic meaning.

Practice

Run meeting_13.do using panel.csv. Compare the pooled and fixed-effects beer-tax slopes. Translate a ten-cent tax difference under the fixed-effects fit. Then compare the endpoint first-difference result and explain why it need not equal the seven-year fixed-effects estimate.

Distinguish within-state and endpoint comparisons: explanation

A reproducible result needs both a calculation and an interpretation.

> The pooled slope with year controls is 0.366336; the state fixed-effects slope with year controls is -0.639980. A 0.10-dollar tax contrast in the latter fit gives -0.063998 fatalities per 10,000 residents. Its uncertainty uses state clusters, reflecting repeated observations within 48 states. The 1982-to-1988 first-difference slope is -1.040973. That fit uses only two endpoints; the seven-year fixed-effects fit uses all within-state changes in its specified sample. The familiar equivalence between first differences and fixed effects concerns compatible two-period models, rather than these different samples. Both methods remove stable state components, while changing state-level confounding can remain.

Next, seek a different source of treatment variation

Panel adjustment may leave time-varying endogeneity.

> Panel methods remove stable unit effects and let us form explicit change comparisons.

> The next lesson introduces instrumental variables, using an outside source of variation that moves the regressor under specified conditions.